Unilateral z-transform and mixed practice

Answers and worked solutions · Day 16 · Sun 8 Nov · pack D9

5. Practice

From your course files

T3 Q3 (tutorial; do (b), redone with $Y = HX$).

Hint: at rest, so $Y = HX$. This is the zero-state part of Example 2.

(a) $\lambda^2 - \frac{3}{4}\lambda + \frac{1}{8} = 0 \Rightarrow \lambda = \frac{1}{2}, \frac{1}{4}$. The modes are 2^{-n} and 4^{-n}.

(b) Transform the LCCDE with zero initial conditions: $H(z) = \dfrac{1 - z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{4}z^{-1}\right)}$, and $X(z) = \dfrac{1}{1 - \frac{1}{3}z^{-1}}$, $|z| > \frac{1}{3}$. So

$$Y(z) = \frac{1 - z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{4}z^{-1}\right)\left(1 - \frac{1}{3}z^{-1}\right)} = \frac{-6}{1 - \frac{1}{2}z^{-1}} + \frac{-9}{1 - \frac{1}{4}z^{-1}} + \frac{16}{1 - \frac{1}{3}z^{-1}}, \qquad |z| > \tfrac{1}{2}.$$

(Cover-up: at $z^{-1} = 2$: $\frac{1-2}{(1-\frac{1}{2})(1-\frac{2}{3})} = -6$; at $z^{-1} = 4$: $\frac{1-4}{(1-2)(1-\frac{4}{3})} = -9$; at $z^{-1} = 3$: $\frac{1-3}{(1-\frac{3}{2})(1-\frac{3}{4})} = 16$.) All three terms are causal:

$$y[n] = \left(-6 \cdot 2^{-n} - 9 \cdot 4^{-n} + 16 \cdot 3^{-n}\right)u[n],$$

which is exactly the official (time-domain) answer.

T3 Q5 (tutorial; do (b) and (c) via $H(z)$).

Hint: take the causal ROC (outside the largest pole). For (c), $H = \frac{1}{(1-z^{-1})^2}$, then use the $(n+1)p^n$ pair with $p = 1$.

(b)

$$H(z) = \frac{1}{1 + 0.2z^{-1} - 0.08z^{-2}} = \frac{1}{(1 - 0.2z^{-1})(1 + 0.4z^{-1})} = \frac{\frac{1}{3}}{1 - 0.2z^{-1}} + \frac{\frac{2}{3}}{1 + 0.4}$$

, ROC $|z| > 0.4$, so

$$h[n] = \left(\tfrac{1}{3}(0.2)^n + \tfrac{2}{3}(-0.4)^n\right)u[n].$$

(c) $H(z) = \dfrac{1}{1 - 2z^{-1} + z^{-2}} = \dfrac{1}{(1 - z^{-1})^2}$, a double pole at $z = 1$, ROC $|z| > 1$. With $(n+1)p^n u[n] \leftrightarrow \dfrac{1}{(1 - pz^{-1})^2}$ and $p = 1$:

$$h[n] = (1+n)u[n].$$

Both agree with the official solutions, which find the same $h[n]$ from the characteristic modes and $h[0]$, $h[1]$.

T7 Q1 (tutorial).

Hint: factor the 3 out of $(3 - z^{-1})$ first, so that both factors read $(1 - pz^{-1})$.

$$X(z) = \frac{\frac{5}{3}z^{-1}}{(1-2z^{-1})\left(1-\frac{1}{3}z^{-1}\right)} = \frac{1}{1-2z^{-1}} - \frac{1}{1-\frac{1}{3}z^{-1}} \qquad [\text{partial fraction expansion}]$$

Two poles at radii $\frac{1}{3}$ and 2 give three possible ROCs, each with its own signal:

$$x(n) = \begin{cases} 2^n u(n) - (\frac{1}{3})^n u(n), & |z| > 2 \text{ (causal, unstable)} \\ -2^n u(-n-1) - (\frac{1}{3})^n u(n), & \frac{1}{3} < |z| < 2 \text{ (two-sided, the absolutely summable one)} \\ -2^n u(-n-1) + (\frac{1}{3})^n u(-n-1), & |z| < \frac{1}{3} \text{ (anti-causal)} \end{cases}$$

The official solution also lists $2^n u(n) + (\frac{1}{3})^n u(-n-1)$ with ROC $\varnothing$. That one is not a valid signal for this $X(z)$ (its two ROCs $|z| > 2$ and $|z| < \frac{1}{3}$ do not overlap), so there are 3 real answers.

T7 Q2 (tutorial).

Hint: $1 - \frac{10}{3}z^{-1} + z^{-2} = (1 - \frac{1}{3}z^{-1})(1 - 3z^{-1})$.

$$X(z) = \frac{3}{\left(1-\frac{1}{3}z^{-1}\right)(1-3z^{-1})} = -\frac{\frac{3}{8}}{1-\frac{1}{3}z^{-1}} + \frac{\frac{27}{8}}{1-3z^{-1}} \qquad [\text{partial fraction expansion}]$$

Absolutely summable $x(n)$ means the ROC of $X(z)$ must include the unit circle $|z| = 1$, so the ROC is $\frac{1}{3} < |z| < 3$. The pole at $\frac{1}{3}$ gives a causal term and the pole at 3 an anti-causal term:

$$x(n) = -\frac{3}{8}\left(\frac{1}{3}\right)^n u(n) - \frac{27}{8}\,3^n u(-n-1), \qquad \text{ROC: } \tfrac{1}{3} < |z| < 3.$$

T7 Q3 (tutorial).

Hint: combine $X(z)$ into one fraction first; then $H = Y/X$.

(a) Using the delay property on the second term of $x(n)$:

$$X(z) = \frac{1}{1-\frac{1}{2}z^{-1}} - \frac{\frac{1}{4}z^{-1}}{1-\frac{1}{2}z^{-1}} = \frac{1-\frac{1}{4}z^{-1}}{1-\frac{1}{2}z^{-1}}, \qquad Y(z) = \frac{1}{1-\frac{1}{3}z^{-1}}.$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 - \frac{1}{2}z^{-1}}{\left(1 - \frac{1}{4}z^{-1}\right)\left(1 - \frac{1}{3}z^{-1}\right)} = \frac{1 - \frac{1}{2}z^{-1}}{1 - \frac{7}{12}z^{-1} + \frac{1}{12}z^{-2}} = \frac{3}{1 - \frac{1}{4}z^{-1}} - \frac{2}{1 - \frac{1}{3}z^{-1}}.$$

As the system is causal, the ROC is $|z| > \frac{1}{3}$ and

$$h(n) = 3\left(\frac{1}{4}\right)^n u(n) - 2\left(\frac{1}{3}\right)^n u(n).$$

(b) Cross-multiply $Y(z)\left(1 - \frac{7}{12}z^{-1} + \frac{1}{12}z^{-2}\right) = X(z)\left(1 - \frac{1}{2}z^{-1}\right)$ and invert term by term:

$$y(n) - \frac{7}{12}y(n-1) + \frac{1}{12}y(n-2) = x(n) - \frac{1}{2}x(n-1).$$

(c) Stable, because the ROC of $H(z)$, $|z| > \frac{1}{3}$, includes the unit circle $|z| = 1$.

All three answers were reproduced in Python (sympy partial fractions) and agree with the official solutions.

New questions

Part A: unilateral z-transform (graded easy to hard)

a. $Y - 0.5\left[z^{-1}Y + 4\right] = \frac{1}{1-z^{-1}}$, so $Y(1 - 0.5z^{-1}) = 2 + \frac{1}{1-z^{-1}}$:

$$Y(z) = \frac{2}{1 - 0.5z^{-1}} + \frac{1}{(1 - z^{-1})(1 - 0.5z^{-1})}$$

c. The first term is the zero-input part: $y_{zi}(n) = 2(0.5)^n u(n)$. For the second term, the residue at $z^{-1} = 1$ is $\frac{1}{0.5} = 2$ and at $z^{-1} = 2$ it is $\frac{1}{1-2} = -1$. So $y_{zs}(n) = \left(2 - (0.5)^n\right)u(n)$.

d. $\boxed{y(n) = \left(2 + (0.5)^n\right)u(n)}$. Check: $y(0) = 1 + 0.5(4) = 3$ (checks), $y(1) = 1 + 1.5 = 2.5$ (checks).

a. $a_1 = -0.7$, $a_2 = 0.1$. $I(z) = -0.7(1) + 0.1\left[z^{-1}(1) + 5\right] = -0.2 + 0.1z^{-1}$. With $A(z) = (1 - 0.5z^{-1})(1 - 0.2z^{-1})$, we get $A(z)Y = 1 + 0.2 - 0.1z^{-1}$.

b. $Y_{zi} = \frac{0.2 - 0.1z^{-1}}{A(z)} = \frac{0.2(1 - 0.5z^{-1})}{(1 - 0.5z^{-1})(1 - 0.2z^{-1})} = \frac{0.2}{1 - 0.2z^{-1}}$. So $y_{zi}(n) = 0.2(0.2)^n u(n)$. The ICs excite only the 0.2^n mode.

c. $Y_{zs} = \frac{1}{A(z)} = H(z)$, because $X(z) = 1$. The residue at $z^{-1} = 2$ is $\frac{1}{1-0.4} = \frac{5}{3}$, and at $z^{-1} = 5$ it is $\frac{1}{1-2.5} = -\frac{2}{3}$. So $y_{zs}(n) = h(n) = \left(\frac{5}{3}(0.5)^n - \frac{2}{3}(0.2)^n\right)u(n)$. The impulse response is defined with zero ICs (L04 slide 13), so it is the zero-state response to $\delta(n)$.

d. $\boxed{y(n) = \left(\frac{5}{3}(0.5)^n - \frac{7}{15}(0.2)^n\right)u(n)}$. The recursion gives $y(0) = 1 + 0.7(1) - 0.1(5) = 1.2$ and $y(1) = 0.7(1.2) - 0.1(1) = 0.74$. The formula gives $\frac{5}{3} - \frac{7}{15} = 1.2$ (checks) and $\frac{5}{6} - \frac{7}{75} = 0.74$ (checks).

Here $a_1 = 0$ and $a_2 = -0.25$. The input term $x(n-1)$ gives $z^{-1}X$ with no extra term, because $x(-1) = 0$.

$I(z) = -0.25\left[z^{-1}(2) + (-4)\right] = 1 - 0.5z^{-1}$. Also
$A(z) = 1 - 0.25z^{-2} = (1 - 0.5z^{-1})(1 + 0.5z^{-1})$.

Zero-input: $Y_{zi} = \frac{-(1-0.5z^{-1})}{(1-0.5z^{-1})(1+0.5z^{-1})} = \frac{-1}{1+0.5z^{-1}}$, so $y_{zi}(n) = -(-0.5)^n u(n)$.

Zero-state: $Y_{zs} = \frac{1+z^{-1}}{(1-0.5z^{-1})(1+0.5z^{-1})(1-\frac{1}{3}z^{-1})}$. The residues are:

- at $z^{-1} = 2$: $\frac{3}{(2)(\frac{1}{3})} = 4.5$
- at $z^{-1} = -2$: $\frac{-1}{(2)(\frac{5}{3})} = -0.3$
- at $z^{-1} = 3$: $\frac{4}{(-0.5)(2.5)} = -3.2$

So $y_{zs}(n) = \left(4.5(0.5)^n - 0.3(-0.5)^n - 3.2(\frac{1}{3})^n\right)u(n)$.

TOTAL:

$$\boxed{y(n) = \left(4.5(0.5)^n - 1.3(-0.5)^n - 3.2(\tfrac{1}{3})^n\right)u(n)}$$

Check: the recursion gives $y(0) = x(0) + x(-1) + 0.25y(-2) = 1 + 0 - 1 = 0$, and the formula gives $4.5 - 1.3 - 3.2 = 0$ (checks). Also $y(1) = \frac{1}{3} + 1 + 0.25(2) = \frac{11}{6}$, and the formula gives $2.25 + 0.65 - 1.0667 = 1.8333$ (checks).

$A(z) = (1 - 0.5z^{-1})^2$, a double pole at 0.5.
$I(z) = -1(2) + 0.25\left[2z^{-1} + 0\right] = -2 + 0.5z^{-1}$.

a. $Y_{zi} = \frac{2-0.5z^{-1}}{(1-0.5z^{-1})^2} = \frac{B}{1-0.5z^{-1}} + \frac{C}{(1-0.5z^{-1})^2}$. Then $C =$ the numerator at $z^{-1} = 2$, which is $2 - 1 = 1$. Setting $z^{-1} = 0$ gives $2 = B + C$, so $B = 1$.

$$y_{zi}(n) = \left[(0.5)^n + (n+1)(0.5)^n\right]u(n) = (n+2)(0.5)^n u(n)$$

b. $Y_{zs} = \frac{1}{(1-z^{-1})(1-0.5z^{-1})^2}$. The residue at $z^{-1} = 1$ is $\frac{1}{0.25} = 4$. C at $z^{-1} = 2$ is $\frac{1}{1-2} = -1$. Setting $z^{-1} = 0$ gives $1 = 4 + B - 1$, so $B = -2$.

$$y_{zs}(n) = \left[4 - 2(0.5)^n - (n+1)(0.5)^n\right]u(n) = \left[4 - (n+3)(0.5)^n\right]u(n)$$

c. $\boxed{y(n) = \left(4 - (0.5)^n\right)u(n)}$. The $n(0.5)^n$ terms cancel exactly. You can see why in the combined form: the numerator $1 + (2 - 0.5z^{-1})(1 - z^{-1})$ factors as $(1 - 0.5z^{-1})(3 - z^{-1})$, which cancels one of the two pole factors. So $Y = \frac{3-z^{-1}}{(1-z^{-1})(1-0.5z^{-1})}$.

d. $y(0) = 1 + 2 - 0 = 3$ and $y(1) = 1 + 3 - 0.25(2) = 3.5$. The formula gives $4 - 1 = 3$ (checks) and $4 - 0.5 = 3.5$ (checks).

a. $H(z) = \frac{1}{1-0.9z^{-1}+0.2z^{-2}} = \frac{1}{(1-0.5z^{-1})(1-0.4z^{-1})}$. The poles are 0.5 and 0.4, so the causal system is stable. The steady state is $H(1) = \frac{1}{0.5 \times 0.6} = \frac{10}{3}$.

b. $I(z) = -0.9c + 0.2cz^{-1}$, so

$$Y(z) = \frac{1 + (0.9c - 0.2cz^{-1})(1 - z^{-1})}{(1 - z^{-1})(1 - 0.5z^{-1})(1 - 0.4z^{-1})}$$

c. Set the numerator to zero at $z^{-1} = 2$: $1 + (0.9c - 0.4c)(1 - 2) = 1 - 0.5c = 0$, so $\boxed{c = y(-1) = 2}$.

d. With $c = 2$, the numerator is $2.8 - 2.2z^{-1} + 0.4z^{-2} = (1 - 0.5z^{-1})(2.8 - 0.8z^{-1})$. So

$$Y = \frac{2.8 - 0.8z^{-1}}{(1 - z^{-1})(1 - 0.4z^{-1})}$$

The residue at $z^{-1} = 1$ is $\frac{2}{0.6} = \frac{10}{3}$, and at $z^{-1} = 2.5$ it is $\frac{0.8}{-1.5} = -\frac{8}{15}$.

$$\boxed{y(n) = \left(\tfrac{10}{3} - \tfrac{8}{15}(0.4)^n\right)u(n)}$$

Check: the recursion gives $y(0) = 1 + 0.9(2) - 0 = 2.8$, and the formula gives $\frac{10}{3} - \frac{8}{15} = 2.8$ (checks).

a. Factor the denominator: $1 - 0.25z^{-1} - 0.125z^{-2} = (1 - 0.5z^{-1})(1 + 0.25z^{-1})$. So

$$H(z) = \frac{1 - z^{-1}}{(1 - 0.5z^{-1})(1 + 0.25z^{-1})} = \frac{z(z - 1)}{(z - 0.5)(z + 0.25)}$$

The zeros are at $z = 0$ and $z = 1$. The poles are at 0.5 and -0.25. For the causal system the ROC is $|z| > 0.5$, which contains the unit circle, so the system is **stable**.

b. $I(z) = -0.25(4) - 0.125\left[4z^{-1} - 8\right] = -1 - 0.5z^{-1} + 1 = -0.5z^{-1}$. So $Y_{zi} = \frac{0.5z^{-1}}{(1-0.5z^{-1})(1+0.25z^{-1})}$.

The residue at $z^{-1} = 2$ is $\frac{1}{1.5} = \frac{2}{3}$, and at $z^{-1} = -4$ it is $\frac{-2}{3} = -\frac{2}{3}$.

$$y_{zi}(n) = \tfrac{2}{3}\left[(0.5)^n - (-0.25)^n\right]u(n)$$

Note $y_{zi}(0) = 0$. This is consistent, because $0.25(4) + 0.125(-8) = 0$.

c. $Y_{zs} = \frac{1-z^{-1}}{(1-0.5z^{-1})^2(1+0.25z^{-1})}$. The input pole at 0.5 lands on a system pole, so there is a double pole and therefore an $(n + 1)(0.5)^n$ term.

- C ($z^{-1} = 2$): $\frac{1-2}{1+0.5} = -\frac{2}{3}$

- A for the (-0.25) term ($z^{-1} = -4$): $\frac{1+4}{(1+2)^2} = \frac{5}{9}$
- B: setting $z^{-1} = 0$ gives $1 = B + C + A$, so $B = \frac{10}{9}$

$$y_{zs}(n) = \left[\tfrac{10}{9}(0.5)^n - \tfrac{2}{3}(n+1)(0.5)^n + \tfrac{5}{9}(-0.25)^n\right]u(n) = \left[\left(\tfrac{4}{9} - \tfrac{2}{3}n\right)(0.5)^n + \tfrac{5}{9}(-0.25)^n\right]u(n)$$

d. $$\boxed{y(n) = \left[\left(\tfrac{10}{9} - \tfrac{2}{3}n\right)(0.5)^n - \tfrac{1}{9}(-0.25)^n\right]u(n)}$$

The recursion gives $y(0) = 1 - 0 + 0.25(4) + 0.125(-8) = 1$, and the formula gives $\frac{10}{9} - \frac{1}{9} = 1$ (checks). Next, $y(1) = 0.5 - 1 + 0.25(1) + 0.125(4) = 0.25$, and the formula gives $\frac{5}{9} - \frac{1}{3} + \frac{1}{36} = \frac{1}{4}$ (checks).

Part B: mixed z-domain set (exam-style, about 12 min each)

a. In powers of z, $X(z) = \frac{z(2z-2.5)}{(z-0.5)(z+2)}$. The zeros are at $z = 0$ and $z = 1.25$. The poles are at $z = 0.5$ and $z = -2$.

b. The residue at $z^{-1} = 2$ is $\frac{2-5}{1+4} = -0.6$, and at $z^{-1} = -0.5$ it is $\frac{2+1.25}{1+0.25} = 2.6$.

$$X(z) = \frac{-0.6}{1 - 0.5z^{-1}} + \frac{2.6}{1 + 2z^{-1}}$$

c.

ROC	$x(n)$	Type
$\lvert z\rvert > 2$	$-0.6(0.5)^n u(n) + 2.6(-2)^n u(n)$	causal
$0.5 < \lvert z\rvert < 2$	$-0.6(0.5)^n u(n) - 2.6(-2)^n u(-n-1)$	two-sided
$\lvert z\rvert < 0.5$	$0.6(0.5)^n u(-n-1) - 2.6(-2)^n u(-n-1)$	anti-causal

d. Only the ring $0.5 < |z| < 2$ contains $|z| = 1$, so only the two-sided signal has a DTFT. $X(\omega)\big|_{\omega=0} = X(1) = \frac{2-2.5}{(0.5)(3)} = -\frac{1}{3}$.

e. $X^+(z)$ keeps only the $n \geq 0$ samples, which here are $-0.6(0.5)^n u(n)$. So $X^+(z) = \frac{-0.6}{1-0.5z^{-1}}$. It differs from $X(z)$ because the signal is not causal. Its left-sided part, $-2.6(-2)^n u(-n-1)$, is thrown away.

Numerical check: summing each sequence against z^{-n} at a test point inside each ROC reproduced $X(z)$ to at least 10 significant figures.

a. Factor the denominator: $1 + 0.1z^{-1} - 0.3z^{-2} = (1 - 0.5z^{-1})(1 + 0.6z^{-1})$. So $H(z) = \frac{1}{(1-0.5z^{-1})(1+0.6z^{-1})}$ with ROC $|z| > 0.6$. The poles are 0.5 and -0.6.

The residue at $z^{-1} = 2$ is $\frac{1}{2.2} = \frac{5}{11}$. At $z^{-1} = -\frac{5}{3}$ it is $\frac{1}{1+\frac{5}{6}} = \frac{6}{11}$.

$$h(n) = \left(\tfrac{5}{11}(0.5)^n + \tfrac{6}{11}(-0.6)^n\right)u(n)$$

b. Yes. The system is causal and both poles are inside $|z| = 1$, so the ROC $|z| > 0.6$ contains the unit circle.

c. $I(z) = 0.1(1) - 0.3\left[z^{-1}(1) + 0\right] = 0.1 - 0.3z^{-1}$.

Zero-input: $Y_{zi} = \frac{-0.1+0.3z^{-1}}{(1-0.5z^{-1})(1+0.6z^{-1})}$. The residue at $z^{-1} = 2$ is $\frac{0.5}{2.2} = \frac{5}{22}$, and at $z^{-1} = -\frac{5}{3}$ it is $\frac{-0.6}{11/6} = -\frac{18}{55}$.

$$y_{zi}(n) = \left(\tfrac{5}{22}(0.5)^n - \tfrac{18}{55}(-0.6)^n\right)u(n)$$

Zero-state: $Y_{zs} = \frac{1}{(1-z^{-1})(1-0.5z^{-1})(1+0.6z^{-1})}$. The residues are:

- at $z^{-1} = 1$: $\frac{1}{0.5\times 1.6} = 1.25$
- at $z^{-1} = 2$: $\frac{1}{(-1)(2.2)} = -\frac{5}{11}$
- at $z^{-1} = -\frac{5}{3}$: $\frac{1}{(\frac{8}{3})(\frac{11}{6})} = \frac{9}{44}$

$$y_{zs}(n) = \left(1.25 - \tfrac{5}{11}(0.5)^n + \tfrac{9}{44}(-0.6)^n\right)u(n)$$

TOTAL:

$$\boxed{y(n) = \left(1.25 - \tfrac{5}{22}(0.5)^n - \tfrac{27}{220}(-0.6)^n\right)u(n)}$$

Check: the recursion gives $y(0) = 1 - 0.1(1) + 0.3(0) = 0.9$, and the formula gives $1.25 - 0.2273 - 0.1227 = 0.9$ (checks).

d. $y(n)$ settles to $1.25 = H(1)$. The zero-input response contains only the system modes $(0.5)^n$ and $(-0.6)^n$, and both decay. For a stable system the ICs therefore affect only the transient.

a. $Y(z) = \frac{4}{1-z^{-1}} - \frac{4}{1-0.5z^{-1}} + \frac{1}{1-0.25z^{-1}}$. Over the common denominator, the numerator is

$$4(1 - 0.75z^{-1} + 0.125z^{-2}) - 4(1 - 1.25z^{-1} + 0.25z^{-2}) + (1 - 1.5z^{-1} + 0.5z^{-2}) = 1$$

Then $H = Y/X = Y(1 - z^{-1})$:

$$H(z) = \frac{1 + 0.5z^{-1}}{(1 - 0.5z^{-1})(1 - 0.25z^{-1})} = \frac{1 + 0.5z^{-1}}{1 - 0.75z^{-1} + 0.125z^{-2}}$$

b. $y(n) - 0.75y(n-1) + 0.125y(n-2) = x(n) + 0.5x(n-1)$.

c. The residue at $z^{-1} = 2$ is $\frac{2}{0.5} = 4$, and at $z^{-1} = 4$ it is $\frac{3}{-1} = -3$. So $h(n) = \left(4(0.5)^n - 3(0.25)^n\right)u(n)$. The system is stable because both poles are inside the unit circle and the ROC is $|z| > 0.5$. Also $H(1) = 4$, which agrees with the step steady state.

d. $I(z) = -0.75(4) + 0.125\left[4z^{-1} + 4\right] = -2.5 + 0.5z^{-1}$. So $Y_{zi} = \frac{2.5 - 0.5z^{-1}}{(1-0.5z^{-1})(1-0.25z^{-1})}$. The residue at $z^{-1} = 2$ is $\frac{1.5}{0.5} = 3$, and at $z^{-1} = 4$ it is $\frac{0.5}{-1} = -0.5$. The zero-state part is the given step response.

$$\boxed{y(n) = \left[4 - (0.5)^n + 0.5(0.25)^n\right]u(n)}, \quad y(0) = 3.5$$

Why it is not constant: a step that starts at $n = 0$ has $x(-1) = 0$. At $n = 0$, the term $0.5x(n-1)$ therefore contributes 0, not 0.5. A true steady state needs the input to have been on for ever, so that $x(-1) = 1$. In that case the recursion gives $y(0) = 1 + 0.5 + 3 - 0.5 = 4$. The missing 0.5 sets off a transient that decays.

a. In powers of z, $H = \frac{1.5z}{(z-0.5)(z-2)}$. The zeros are at $z = 0$ and $z = \infty$ (because $H(\infty) = 0$). The poles are at 0.5 and 2. That is two poles and two zeros, as L07 slide 17 requires.

b. The residue at $z^{-1} = 0.5$ is $\frac{0.75}{0.75} = 1$, and at $z^{-1} = 2$ it is $\frac{3}{-3} = -1$. So $H = \frac{1}{1-2z^{-1}} - \frac{1}{1-0.5z^{-1}}$.

ROC	$h(n)$	Causal?	Stable?
$\lvert z\rvert > 2$	$(2^n - 0.5^n)u(n)$	yes	no ($\lvert z\rvert = 1$ is not in the ROC)
$0.5 < \lvert z\rvert < 2$	$-2^n u(-n-1) - 0.5^n u(n)$	no (two-sided)	yes
$\lvert z\rvert < 0.5$	$-2^n u(-n-1) + 0.5^n u(-n-1)$	no (anti-causal)	no

c. $y(n) - 2.5y(n-1) + y(n-2) = 1.5x(n-1)$.

d. $Y = \frac{1.5z^{-1}}{(1-z^{-1})(1-0.5z^{-1})(1-2z^{-1})}$. The residues are 2 at the pole 2, -3 at the pole 1 and 1 at the pole 0.5. So $y(n) = \left(2 \cdot 2^n - 3 + 0.5^n\right)u(n)$, which is unbounded.

e. Let $y(-1) = c$. Then $I(z) = -2.5c + cz^{-1}$, and

$$Y = \frac{1.5z^{-1} + (2.5c - cz^{-1})(1 - z^{-1})}{(1 - z^{-1})(1 - 0.5z^{-1})(1 - 2z^{-1})}$$

To remove the 2^n mode, the numerator must be zero at $z^{-1} = 0.5$: $0.75 + (2c)(0.5) = 0$, so $\boxed{c = y(-1) = -0.75}$. The numerator then factors as $(1 - 2z^{-1})(-1.875 + 0.375z^{-1})$. That leaves $Y = \frac{-1.875 + 0.375z^{-1}}{(1-z^{-1})(1-0.5z^{-1})}$, and

$$\boxed{y(n) = \left(-3 + 1.125(0.5)^n\right)u(n)}$$

Check: the recursion gives $y(0) = 0 + 2.5(-0.75) - 0 = -1.875$, and the formula gives $-3 + 1.125 = -1.875$ (checks).

f. The cancellation needs $y(-1)$ to be exactly -0.75. In simulation, $y(-1) = -0.7501$ gives $y(20) \approx -283$ and $y(39) \approx -1.5 \times 10^8$. Any noise or rounding re-excites the 2^n mode. Stability is a property of $H(z)$ and its ROC (L08 slide 8), not of the initial conditions.

6. Quick recall check

Question

1. $z^{-2}Y(z) + z^{-1}y(-1) + y(-2)$.
2. $\frac{z^{-1}}{1-z^{-1}}$. There is no extra term because $x(-1) = 0$.
3. The zero-state part. The zero-input part contains only the system poles (the characteristic modes), never the input poles.
4. $(n+1)p^n u(n)$.
5. Set $z^{-1} = 0$: $y(0) = 3$ (initial value theorem; this is L08's slide example).